

SP2 Requirements – Orders and Reports

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Pages: Orders (ORD) and Reports & Audit (RPT)

1. Stakeholders

Actor	Use cases
IT administrators	Review reports, approve orders, monitor audits and monthly summaries
IT student assistants	Create daily reports, track orders, update order status when items arrive
People recorded in the system who do not use it	Faculty and staff who appear in records (for example, as requesters or room users) but never log in

2. Requirement-Gathering Methods

- **Document review:** examine existing forms, tickets, and documents used for reports and orders.
- **Questionnaires:** collect common needs for a tracking system from IT department staff.
- **Use-case discussions:** describe typical interactions between users and the system.

3. Questions

Reports & Audit

- What statuses can a report have?
- What components does a report contain, and which reports are required?
- Why do we need each report?
- What format should reports use? Should they be editable after submission?
- What is the next step after a report is received?

Orders

- What statuses can an order have?
- When an order arrives, what is the next step?
- When should an order be flagged? What event triggers an automatic order?

4. Functional Requirements

Reports & Audit

ID	Requirement	Priority
FR-RPT-01	The system shall allow users to create new reports.	Must Have
FR-RPT-02	The system shall allow users to group reports by project.	Should Have
FR-RPT-03	The system shall notify the assigned reviewers when a report is submitted.	Must Have
FR-RPT-04	The system shall allow users to change a report's status (Submitted, Reviewed, Rejected, Passed).	Must Have
FR-RPT-05	The system shall provide templates for daily reports, such as the morning report completed by the student assistant on the morning shift.	Should Have
FR-RPT-06	The system shall keep an audit record of all tracked activities, including tasks, orders, and their progress, not only reports.	Must Have
FR-RPT-07	The system shall automatically send a monthly summary report.	Should Have
FR-RPT-08	The system shall allow users to search and filter reports and audit records.	Must Have
FR-RPT-09	The system shall assign a unique ID to every report and audit record.	Must Have

Orders

ID	Requirement	Priority
FR-ORD-01	The system shall link orders to the inventory and automatically flag items that are almost out of stock.	Must Have
FR-ORD-02	The system shall track every order and its status.	Must Have
FR-ORD-03	When an order's status changes to Arrived, the system shall automatically add the received quantity to the inventory.	Must Have
FR-ORD-04	The system shall automatically summarize orders.	Should Have
FR-ORD-05	The system shall allow users to search and filter orders.	Must Have
FR-ORD-06	The system shall assign a unique ID to every order.	Must Have

Almost out-of-stock here should be understood as when the quantity of an item is less than a threshold; it will be automatically added to the order and send a notification through Gmail and in notification pages.

5. Non-Functional Requirements

These apply to both pages.

ID	Requirement	Priority
NFR-RPT-01	The system shall be easy to use and understand.	Must Have
NFR-RPT-02	The system shall store all data within its own database, and the data shall remain available after the program is closed.	Must Have

ID	Requirement	Priority
NFR-RPT-03	The system shall display a clear error message for invalid input.	Must Have
NFR-RPT-04	The design shall follow object-oriented principles.	Must Have

6. Business Rules

ID	Business rule
BR-ORD-01	An item cannot have more than one open order at the same time.
BR-RPT-01	The monthly summary is sent only at the end of each month.
BR-RPT-02	Only IT administrators and IT student assistants can add, update, or modify reports and orders.
BR-RPT-03	Every audit record has a status, and audit records are never removed from the system.

7. Main Use Cases

ID	Use case	Primary actor	Result
UC-ORD-01	Create Order	IT student assistant	A new order is recorded with status Requested
UC-ORD-02	Update Order Status	IT student assistant	The order moves to its next status
UC-ORD-03	Receive Order	IT student assistant	The order is marked Arrived and the inventory quantity increases
UC-ORD-04	Flag Low Stock	System	Items almost out of stock are flagged for ordering

ID	Use case	Primary actor	Result
UC-RPT-01	Create Report	IT student assistant	A new report is saved, optionally from a daily template
UC-RPT-02	Review Report	IT administrator	The report status changes to Reviewed, Rejected, or Passed
UC-RPT-03	Send Monthly Summary	System	A summary of reports and orders is sent at month end
UC-RPT-04	View Audit History	IT administrator or student assistant	Audit records matching the search or filter are displayed

8. Use-Case Specification: Receive Order

Use case ID: UC-ORD-03

Primary actor: IT student assistant

Precondition: The user is logged in, and the order exists with status Ordered.

Main flow:

1. The user opens the Orders page and searches for the order by ID or item.
2. The system displays the order details: item, quantity ordered, vendor, and status.
3. The user selects "Receive Order."
4. The user enters the quantity received.
5. The system validates the quantity.
6. The system changes the order status to Arrived and records the arrival date and the user.
7. The system adds the received quantity to the item's stock in the inventory.
8. The system clears the item's low-stock flag if the new quantity is above the threshold.
9. The system creates an audit record of the change.
10. The system displays a confirmation message.

Alternative flows:

- **1a.** The order ID does not exist: the system displays an error.
- **2a.** The order status is not Ordered (for example, already Arrived or Cancelled): the system rejects the action and shows the current status.
- **4a.** The quantity received is less than the quantity ordered: the system records a partial delivery, adds the received quantity, and keeps the order open for the rest.

- **5a.** The quantity is invalid (zero, negative, or not a number): the system displays an error and asks again.

Postcondition:

- The order status is Arrived (or Partially Arrived).
- The inventory quantity is updated.
- An audit record of the change is saved.

9. Proposed Object-Oriented Classes

Class	Main responsibilities
Order	Store item, quantity ordered and received, vendor, requester, dates, and status
OrderStatus (enum)	Requested, Ordered, Partially Arrived, Arrived, Cancelled
OrderService	Create orders, update status, receive orders, summarize orders, prevent duplicate open orders
StockMonitor	Watch inventory quantities and flag items that fall below their threshold
Report (abstract)	Common report data: ID, title, author, project, status, created date
DailyReport	A report filled in by a student assistant, such as the morning report
MonthlySummaryReport	A report generated by the system at month end from reports and orders
ReportTemplate	A reusable checklist used to create daily reports
ReportStatus (enum)	Submitted, Reviewed, Rejected, Passed
ReportService	Create, group, review, and search reports; send the monthly summary
AuditRecord	Store one change: ID, user, action, entity, old and new values, timestamp, status
AuditLog	Add and search audit records; never delete them

Possible relationships:

- An Order refers to one inventory item (owned by the Assets page).
- OrderService manages many Order objects and updates the inventory when an order arrives.
- StockMonitor observes inventory items and notifies OrderService when stock is low (Observer pattern).
- DailyReport and MonthlySummaryReport extend Report (inheritance and polymorphism).
- A DailyReport is created from one ReportTemplate.
- A Report belongs to one project and has one or more reviewers.
- AuditLog contains many AuditRecord objects; OrderService and ReportService write to it on every change.

10. Acceptance Criteria

Requirement: Receive an order (FR-ORD-03)

The requirement is accepted when:

- An order with status Ordered can be selected.
- Entering the quantity received changes the status to Arrived.
- The item's inventory quantity increases by exactly the quantity received.
- A partial delivery keeps the order open with status Partially Arrived.
- Invalid quantities (zero, negative, non-numeric) are rejected with an error message.
- An audit record is created for the change.

Requirement: Flag low stock (FR-ORD-01)

The requirement is accepted when:

- An item whose quantity falls below its threshold is flagged.
- An item that already has an open order is not flagged for a second order (BR-ORD-01).
- The flag is cleared when stock rises above the threshold.

Requirement: Change a report's status (FR-RPT-04)

The requirement is accepted when:

- A reviewer can set a submitted report to Reviewed, Rejected, or Passed.
- The new status is saved and shown on the report.
- The report's author is notified of the change.
- An audit record is created with the old and new status.

Requirement: Send the monthly summary (FR-RPT-07)

The requirement is accepted when:

- The summary is generated only at the end of the month (BR-RPT-01).
- It includes that month's reports, orders, and their statuses.
- It is delivered to IT administrators.

Requirement: Keep audit records (FR-RPT-06)

The requirement is accepted when:

- Every change to a report or order creates an audit record.
- Each audit record has an ID, user, timestamp, and status.
- No user can delete an audit record (BR-RPT-03).

11. Out-of-Scope Requirements

The following are excluded from this version of the Orders and Reports pages:

- Online payment or vendor invoicing
- Placing orders directly with vendors or through vendor portals (the system only flags and tracks orders)
- Budget and accounting management
- Email or SMS delivery of notifications and summaries (delivered inside the system only)
- Barcode or RFID scanning when receiving orders
- Editing or deleting audit records
- Mobile application